SDET COURSE PROJECT 13 · STUDY GUIDE

Threshold Effects in Time-Delay Tracking

CRLB vs Ziv–Zakai bound for a delay-locked loop and a batch maximum-likelihood estimator of BPSK signals

This guide explains **every concept from first principles**, gives **complete derivations and proofs**, records **every design decision and why it was made**, presents **all results with their uncertainty**, summarises the **four literature surveys**, lays out the **proposal for the remaining 30%**, and ends with a **question bank with follow-ups**.

Prepared for the 70% evaluation, 5–6 November 2026. Repository: github.com/nilavra17ghosh/TIME-DELAY-TRACKING. Every number is traceable to an executed notebook (notebooks 00–07).

DONE completed and verified **PARTIAL** partly done **PROPOSED** not yet implemented

CONTENTS

0 How to use this guide
1 The problem in one picture
2 Mathematical toolkit (with proofs)
3 The signal model and why each choice was made
4 Sprint 1 — the batch MLE
5 Sprint 3 — the Cramér–Rao lower bound
6 The threshold effect
7 Sprint 4 — the Ziv–Zakai bound
8 Sprint 2 — the delay-locked loop
9 Experimental methodology and validation
10 Results (Sprints 1–6)
11 Decision log and mistakes we fixed
12 What the four surveys taught us
13 Limitations
14 Proposal for the remaining 30%
15 Question bank with follow-ups
A Notation and equation sheet

0 How to use this guide

Read Sections 1–8 with a pen: every derivation is written so you can redo it on paper. Sections 9–13 are what you show and defend. Section 14 is the research proposal. Section 15 is for rehearsal: cover the answer, say it aloud, then check. The notebooks contain the same mathematics next to the code that implements it.

Item	Status	Where
Signal model, checks	DONE	notebook 00
Batch MLE, error vs SNR, error distributions	DONE	notebook 01
DLL: S-curve, pull-in, tracking, jitter	DONE	notebook 02
CRLB derivation, checks, overlay	DONE	notebook 03
ZZB derivation, limits, numerical ZZB	DONE	notebook 04
Four-curve comparison, MMSE, DLL theory, DLL failure modes	DONE	notebook 05
Precise thresholds (CIs, 5 pilots), DLL vs MLE	DONE	notebook 06
Prior-width study (T_a = 1, 4, 16)	DONE	notebook 07
Explaining the pilot-to-pilot spread	PARTIAL	measured, not explained
Report and final slides	PARTIAL	evaluation slides only
Dynamic decomposition (research extension)	PROPOSED	Section 14

PART I — FOUNDATIONS

1 The problem in one picture

A transmitter sends a known waveform $s(t)$. The receiver observes it **delayed by an unknown** τ and corrupted by noise:

$$r(t) = s(t - \tau) + w(t).$$

Estimating τ is the core of GNSS ranging, radar, sonar and synchronisation: a delay error of one microsecond is 300 m of range error. The project compares two estimators and two bounds:

	What it is	Key property
Batch MLE	looks at a whole block of data and picks the delay whose replica fits best	optimal at high SNR; needs a search over delays
DLL	a feedback loop that nudges its estimate using "early minus late" correlations	cheap, real-time; can lose lock
CRLB	lower bound on the variance of unbiased estimators	local: only uses the curvature of the true correlation peak
ZZB	lower bound on the MSE of any estimator, built from binary detection	global: sees distant wrong peaks and the prior

The phenomenon — the threshold effect. At high SNR the error is small and follows the CRLB. Below a certain SNR, noise occasionally makes a *wrong* delay look best; the estimate jumps far away and the mean-squared error rises far faster than the CRLB predicts. The project asks: *where is this threshold, does the ZZB predict it, and why do the DLL and the MLE behave differently?*

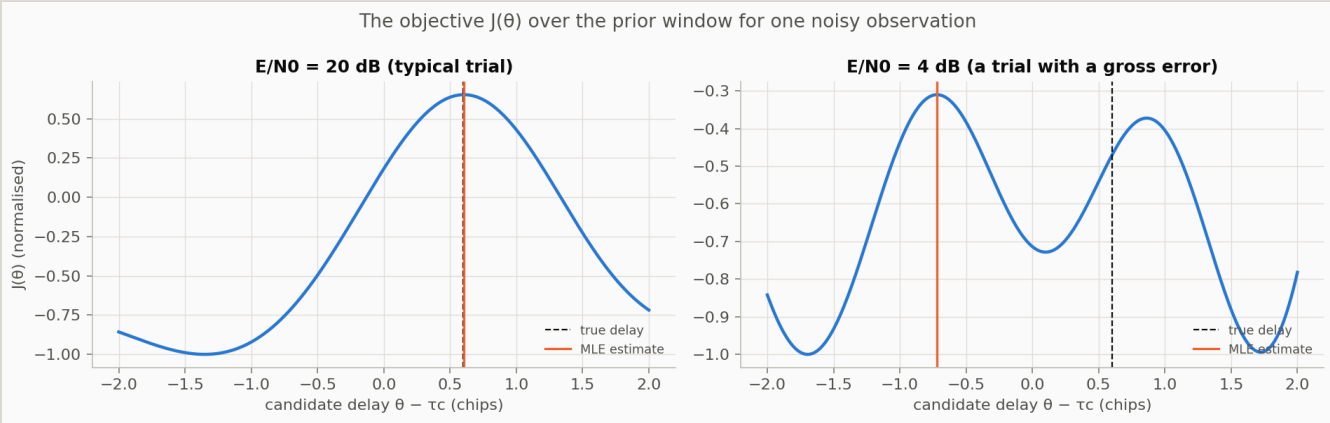


Figure 1. The MLE objective over the search window for one observation. Left (20 dB): one clear peak at the true delay. Right (4 dB, a trial selected to show the mechanism): a noise-induced bump 1.3 chips away is higher than the true peak, so the estimator jumps there. This jump is the threshold effect.

2 Mathematical toolkit (with proofs)

Everything later rests on six facts. Each is proved here so nothing is taken on trust.

2.1 Bias, variance and mean-squared error

For an estimator $\hat{\tau}$ of τ with error $e = \hat{\tau} - \tau$:

$$\underbrace{\mathbb{E}[e^2]}_{\text{MSE}} = \underbrace{\operatorname{var}(\hat{\tau})}_{\text{spread}} + \underbrace{\big(\mathbb{E}[\hat{\tau}] - \tau\big)^2}_{\text{bias}^2}.$$

PROOF

Write $e = (\hat{\tau} - \mathbb{E}\hat{\tau}) + (\mathbb{E}\hat{\tau} - \tau)$. Square and take expectations; the cross term is $2(\mathbb{E}\hat{\tau} - \tau)\,\mathbb{E}[\hat{\tau} - \mathbb{E}\hat{\tau}] = 0$. ∎

WHY IT MATTERS HERE

Below threshold our estimators become **biased** (a delay near the edge of the window is pulled inward). The CRLB bounds only the variance of *unbiased* estimators, while the ZZB bounds the MSE. So all our plots use MSE.

2.2 The Gaussian likelihood

If $r[n] = \mu_n(\theta) + w[n]$ with $w[n] \sim \mathcal{N}(0, \sigma^2)$ independent, the likelihood of the data is

$$p(\mathbf{r};\theta) = \prod_n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\!\Big(-\frac{(r[n]-\mu_n(\theta))^2}{2\sigma^2}\Big), \qquad \ln p(\mathbf{r};\theta) = \text{const} - \frac{1}{2\sigma^2}\sum_n \big(r[n]-\mu_n(\theta$$

Maximising the likelihood is therefore the same as **minimising the squared distance** between the data and the model — least squares.

2.3 The Q-function and binary detection in Gaussian noise

$Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\,du = P(Z > x)$ for $Z \sim \mathcal{N}(0,1)$. Properties used later: $Q(0) = \tfrac{1}{2}$, Q decreases, $Q(x) \le \tfrac{1}{2}e^{-x^2/2}$ for $x \ge 0$.

RESULT (MINIMUM ERROR PROBABILITY BETWEEN TWO KNOWN SIGNALS)

Observe $\mathbf{r} = \mathbf{s}_i + \mathbf{w}$, $i \in \{0,1\}$, equally likely, $\mathbf{w} \sim \mathcal{N}(0, \sigma^2 I)$. The smallest possible probability of deciding wrongly is

$$P_{\min} = Q\!\left(\frac{\|\mathbf{s}_1 - \mathbf{s}_0\|}{2\sigma}\right).$$

PROOF

With equal priors the optimal (MAP) rule picks the hypothesis with the larger likelihood, i.e. the signal closer to $\mathbf{r}$: decide 1 iff $\|\mathbf{r}-\mathbf{s}_1\|^2 < \|\mathbf{r}-\mathbf{s}_0\|^2$. Expanding, this is $T = \langle \mathbf{r}, \mathbf{s}_1 - \mathbf{s}_0\rangle > \tfrac{1}{2}(\|\mathbf{s}_1\|^2 - \|\mathbf{s}_0\|^2)$. Under hypothesis 0, T is Gaussian with mean $\langle \mathbf{s}_0, \mathbf{s}_1 - \mathbf{s}_0\rangle$ and standard deviation $\sigma\|\mathbf{s}_1 - \mathbf{s}_0\|$. The distance from this mean to the threshold is $\tfrac{1}{2}\|\mathbf{s}_1 - \mathbf{s}_0\|^2$ (expand both). So the error probability under 0 is $Q\big(\tfrac{1}{2}\|\mathbf{s}_1-\mathbf{s}_0\|^2/(\sigma\|\mathbf{s}_1-\mathbf{s}_0\|)\big) = Q(\|\mathbf{s}_1-\mathbf{s}_0\|/2\sigma)$, and by symmetry the same under 1. ∎

2.4 Fisher information and the Cramér–Rao inequality

The **score** is $S(\theta) = \partial_\theta \ln p(\mathbf{r};\theta)$, and the **Fisher information** is $I(\theta) = \mathbb{E}[S(\theta)^2]$.

> **THEOREM (CRLB)**
> If $\mathbb{E}[S(\theta)] = 0$ (regularity) and $\hat\theta$ is unbiased, then $\mathrm{var}(\hat\theta) \geq 1/I(\theta)$.

> **PROOF**
> Unbiasedness: $\int \hat\theta(\mathbf{r})\, p(\mathbf{r};\theta)\, d\mathbf{r} = \theta$. Differentiate in θ: $\int \hat\theta\, \partial_\theta p\, d\mathbf{r} = 1$, i.e. $\mathbb{E}[\hat\theta S] = 1$. Because $\mathbb{E}[S] = 0$, also $\mathbb{E}[(\hat\theta - \theta)S] = 1$. By Cauchy–Schwarz, $1 = \mathbb{E}[(\hat\theta-\theta)S]^2 \leq \mathbb{E}[(\hat\theta - \theta)^2]\,\mathbb{E}[S^2] = \mathrm{var}(\hat\theta)\, I(\theta)$. ∎
>
> Equality needs S proportional to $\hat\theta - \theta$: then the estimator is *efficient*. The MLE becomes efficient as the SNR grows (asymptotic efficiency).

2.5 Autocorrelation, bandwidth and Parseval

The normalised autocorrelation of the waveform is $\rho(h) = \sum_n s(t_n)s(t_n - h) \big/ \sum_n s(t_n)^2$, with $\rho(0) = 1$ and $|\rho(h)| \leq 1$ (Cauchy–Schwarz). The **mean-square (Gabor) bandwidth** is

$$\beta^2 = \frac{\sum_n s'(t_n)^2}{\sum_n s(t_n)^2} = \frac{\int (2\pi f)^2 |S(f)|^2 df}{\int |S(f)|^2 df}.$$

The second equality is Parseval's theorem applied to s', whose spectrum is $j2\pi f\, S(f)$. Two facts used later:

> **FACT: $P''(0) = -B^2$**
> $\frac{d^2}{dh^2} \sum_n s(t_n)s(t_n - h)\Big|_{h=0} = \sum_n s(t_n)s''(t_n)$. Summation by parts (exact for a band-limited signal sampled above Nyquist, where sums equal integrals times f_s) gives $\sum s\, s'' = -\sum s'^2$. Dividing by $\sum s^2$: $\rho''(0) = -\beta^2$, hence $1 - \rho(h) \approx \frac{1}{2}\beta^2 h^2$ for small h. ∎

> **A SUBTLETY WE HIT**
> Summation by parts needs the waveform's slope to be continuous. Truncating the pulse with a hard cut left a tiny slope jump; samples landing on it made the measured curvature disagree with β^2. We fixed it by tapering the pulse smoothly over its last chip (Section 11).

2.6 A tail-integral identity

For a non-negative random variable X: $\mathbb{E}[X^2] = \int_0^\infty 2x\, P(X \geq x)\, dx$.

> **PROOF**
> $X^2 = \int_0^X 2x\, dx = \int_0^\infty 2x\, \mathbf{1}\{X \geq x\}\, dx$. Take expectations and swap expectation and integral (allowed because the integrand is non-negative — Tonelli's theorem): $\mathbb{E}[X^2] = \int_0^\infty 2x\, P(X \geq x)\, dx$. ∎

This identity is the first step of the Ziv–Zakai bound: it turns the MSE into a sum over error *sizes*.

3 The signal model and why each choice was made

Notebook 00. Every estimator and bound uses exactly this model, so theory and simulation always describe the same thing.

3.1 The model

$$s(t) = \sum_{k=0}^{N_c-1} a_k\, p(t-k), \qquad r[n] = s(t_n - \tau) + w[n], \quad t_n = \tfrac{n}{4}, \quad w[n] \sim \mathcal{N}(0, \sigma^2),$$

with a known pilot $a_k \in \{\pm 1\}$ of $N_c = 64$ chips, time in units of chips ($T_c = 1$), and the raised-cosine pulse

$$p(t) = \operatorname{sinc}(t)\,\frac{\cos(\pi\alpha t)}{1 - (2\alpha t)^2}, \qquad \alpha = 0.5,$$

smoothly truncated to $|t| \le 6$ chips. p is a **Nyquist pulse**: $p(0) = 1$ and $p(k) = 0$ at every other integer, so at chip centres $s(k) = a_k$. The delay has a uniform prior $\tau \sim \mathcal{U}[\tau_c - T_a/2, \tau_c + T_a/2]$ with $T_a = 4$ chips.

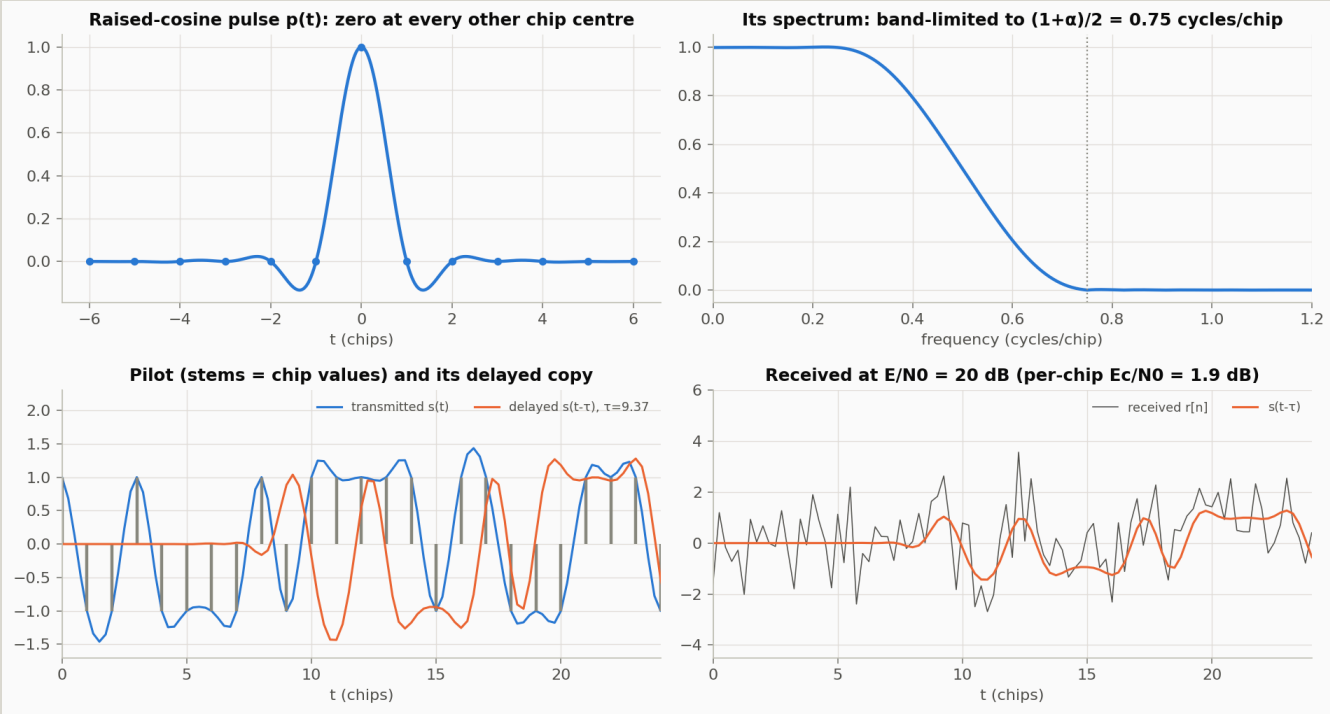


Figure 2. The pulse and its spectrum (band-limited to 0.75 cycles/chip), the pilot with its delayed copy, and what the receiver sees at $E/N_0 = 20$ dB.

3.2 The SNR — derivation

Continuous white noise has two-sided spectral density $N_0/2$. Sampling at rate f_s after an ideal band-limit gives samples of variance $\sigma^2 = \frac{N_0}{2} f_s$. The signal energy is $E = \int s^2 dt = T_s \sum_n s[n]^2$ (exact for a band-limited signal sampled above Nyquist). Therefore

$$\frac{E}{N_0} = \frac{T_s \sum_n s[n]^2}{2\sigma^2 T_s} = \frac{\sum_n s[n]^2}{2\sigma^2} \quad \Longrightarrow \quad \sigma^2 = \frac{\sum_n s[n]^2}{2\,E/N_0}.$$

This is how the code sets the noise for a requested E/N_0. All plots use the **total batch** E/N_0 of the 64-chip pilot; per chip, $E_c/N_0 = E/N_0 - 10\log_{10} 64 = E/N_0 - 18.1$ dB. Check: requesting 10 gives a

measured 10.03.

3.3 Decisions and their reasons

Decision	Reason	What goes wrong otherwise
Raised-cosine pulse	band-limited like a real receiver; analytic spectrum	rectangular chips have infinite bandwidth ⇒ $\beta^2 = \infty$ ⇒ CRLB = 0
4 samples per chip	spectrum ends at 0.75 cycles/chip < Nyquist (2)	aliasing; sums ≠ integrals
Delay applied analytically	$s(t_n - \tau)$ evaluated exactly	interpolation error would contaminate high-SNR results
Known pilot, coherent, real baseband, AWGN	closed-form likelihood, CRLB and ZZB	unknown bits/phase add nuisance parameters (future work)
Uniform prior, $T_a = 4$ chips	the ZZB is Bayesian and needs a prior; wide enough to contain wrong peaks	$T_a = 1$ masks the threshold (Section 10.7)
SNR = batch E/N_0	both bounds depend on total energy	per-sample SNR hides the dependence on pilot length
MSE as the metric	estimators are biased below threshold; ZZB bounds MSE	variance would hide the bias

> **WHY THE MODEL CHANGED AFTER THE FIRST SPRINT-1 SCRIPT**
> The original script (archived) used a 127-chip m-sequence, a Gaussian filter, a fixed delay and per-sample SNR, and showed the same qualitative result. It was replaced because the ZZB needs a *random* delay with a prior, the bounds are naturally written in E/N_0, and the raised cosine makes the CRLB checkable analytically.

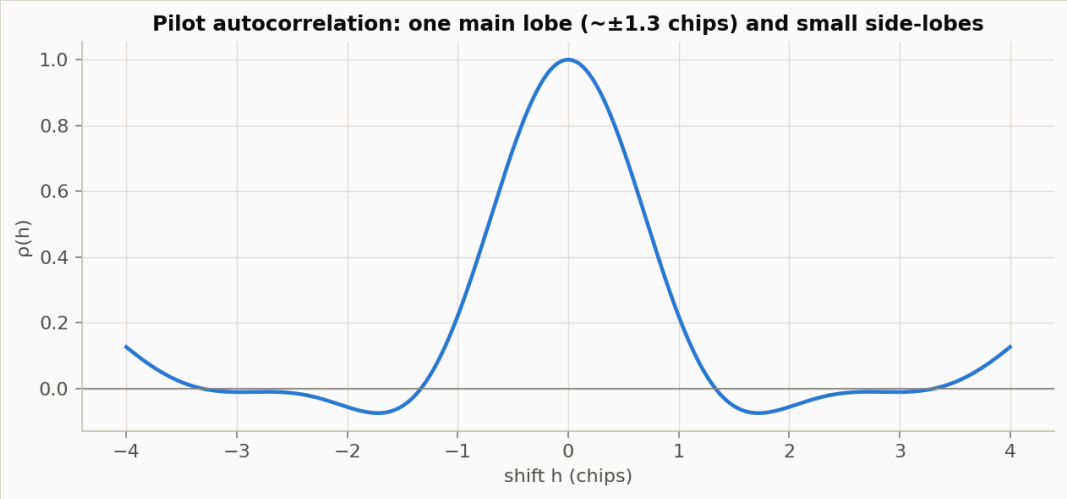


Figure 3. The pilot's autocorrelation $\rho(h)$: main lobe to about ±1.34 chips, side-lobes at most 0.13 beyond 1.5 chips. Its curvature at $h = 0$ sets the CRLB; its side-lobes and the prior width set the ZZB.

4 Sprint 1 — the batch maximum-likelihood estimator

Notebook 01. **DONE**

4.1 Derivation: the MLE is a correlator

With $\mu_n(\theta) = s(t_n - \theta)$, the Gaussian log-likelihood (Section 2.2) expands as

$$\ln p(\mathbf{r};\theta) = \text{const} - \frac{1}{2\sigma^2}\sum_n r[n]^2 + \frac{1}{\sigma^2}\Big(\sum_n r[n]\, s(t_n-\theta) - \tfrac{1}{2}\sum_n s(t_n-\theta)^2\Big).$$

The first sum does not depend on θ, hence

RESULT

$$\hat{\tau}_{\mathrm{ML}} = \arg\max_{\theta\in\text{window}} J(\theta), \qquad J(\theta) = \underbrace{\sum_n r[n]\, s(t_n-\theta)}_{\text{correlation with a shifted replica}} - \tfrac{1}{2}\underbrace{\sum_n s(t_n-\theta)^2}_{\text{replica energy}}.$$

- **Interpretation:** slide a clean replica over the data and pick the best match — a *matched filter*.
- **No SNR needed:** σ^2 multiplies the whole function, so it does not move the arg-max.
- **Energy term:** almost constant because the whole pulse train stays inside the observation window; kept so the estimator is exactly the MLE.
- **Window constraint:** the search is restricted to the prior window. With a uniform prior, the MAP estimator equals this constrained MLE.

4.2 Sub-grid refinement — derivation

J is computed on a grid of step $\Delta = 1/64$ chip. Fit a parabola through the best grid point g and its neighbours: $J(\theta_g + x\Delta) \approx ax^2 + bx + c$ with $J_{g\pm1} = a \pm b + c$, $J_g = c$. So $a = \tfrac{1}{2}(J_{g-1} - 2J_g + J_{g+1})$ and $b = \tfrac{1}{2}(J_{g+1} - J_{g-1})$. The vertex $x^\star = -b/2a$ gives

$$\hat{\theta} = \theta_g + \frac{\Delta}{2}\,\frac{J_{g-1} - J_{g+1}}{J_{g-1} - 2J_g + J_{g+1}}.$$

Check: without noise the error is 0.02 % of the CRLB at 30 dB — negligible.

4.3 The MMSE estimator (posterior mean)

With the uniform prior, Bayes' rule gives $p(\theta \mid \mathbf{r}) \propto \exp\big(J(\theta)/\sigma^2\big)$ on the window, and the estimator with the smallest possible MSE is the posterior mean

$$\hat{\tau}_{\mathrm{MMSE}} = \frac{\int \theta\, e^{J(\theta)/\sigma^2}\, d\theta}{\int e^{J(\theta)/\sigma^2}\, d\theta}.$$

WHY THE POSTERIOR MEAN MINIMISES MSE

For any estimator $g(\mathbf{r})$, $\mathbb{E}[(\tau - g)^2 \mid \mathbf{r}] = \mathrm{var}(\tau \mid \mathbf{r}) + (\mathbb{E}[\tau \mid \mathbf{r}] - g)^2$, minimised by $g = \mathbb{E}[\tau \mid \mathbf{r}]$; averaging over $\mathbf{r}$ keeps the minimum. ■

It is computed from the same grid of J values (no extra cost) but needs σ^2. It matters because the ZZB bounds *every* estimator, so the fair comparison for the bound is the best one (Section 10.6).

5 Sprint 3 — the Cramér–Rao lower bound for our model

Notebook 03. **DONE**

5.1 Derivation in four steps

1. **Score:** $\partial_\tau \ln p = \frac{1}{\sigma^2}\sum_n (r[n] - \mu_n)\,\mu_n' = \frac{1}{\sigma^2}\sum_n w[n]\,\mu_n'(\tau)$. Its mean is 0 because $\mathbb{E}\,w = 0$ (regularity holds).
2. **Fisher information:** $I = \mathbb{E}[(\partial_\tau \ln p)^2] = \frac{1}{\sigma^4}\sum_n\sum_m \mathbb{E}[w_n w_m]\mu_n'\mu_m' = \frac{1}{\sigma^2}\sum_n \mu_n'^2$ (independence: $\mathbb{E}[w_n w_m] = \sigma^2\delta_{nm}$).
3. **Delay model:** $\mu_n'(\tau) = \partial_\tau s(t_n - \tau) = -s'(t_n - \tau)$, so $I = \frac{1}{\sigma^2}\sum_n s'(t_n - \tau)^2$.
4. **Physical form:** substitute $\sigma^2 = \sum s^2/(2E/N_0)$:

RESULT

$$\mathrm{var}(\hat\tau) \;\geq\; \mathrm{CRLB} = \frac{1}{2\,(E/N_0)\,\beta^2} = \frac{1}{2\,N_c\,(E_c/N_0)\,\beta^2}.$$

- **Energy:** doubling the pilot length or the SNR halves the CRLB.
- **Bandwidth:** steeper edges (larger β^2) give a sharper correlation peak and a lower CRLB. Rectangular chips give $\beta^2 = \infty$, which is why we band-limit.
- **Independent of τ** as long as the pulse train stays inside the window (verified: β^2 varies by 10^{-5} across delays).

5.2 Checks

β^2 computed from the spectrum (FFT at 64 samples/chip) is 2.4470; from the samples (the code) 2.4481 — agreement to 0.05 %. For the isolated pulse $\beta^2 = 2.838$; averaged over 400 random pilots, 2.873.

5.3 A finding: the CRLB depends on the pilot

β^2 is the bandwidth of the actual chip pattern: more sign changes mean more high-frequency energy. Over 400 random 64-chip pilots, β^2 ranges from 1.77 to 4.27 — a factor of 2.4 in the CRLB. Consequence: every estimator and bound must use the **same** pilot, and thresholds defined relative to the CRLB move from pilot to pilot.

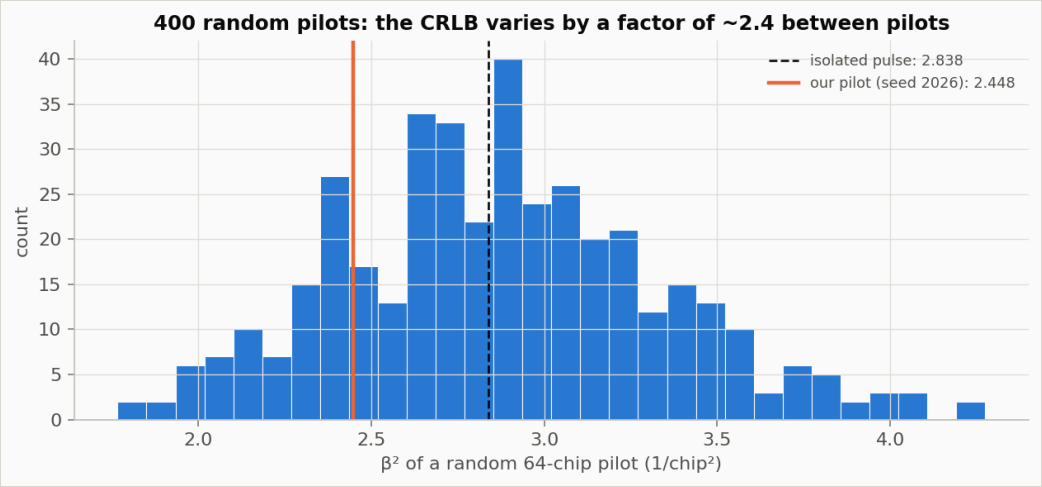


Figure 4. β^2 for 400 random pilots; our pilot (seed 2026) has 2.448.

5.4 Where the CRLB stops making sense — three reasons

1. **It assumes an unbiased estimator.** For a delay fixed 0.5 chip from the edge, the MLE's bias is −1.18 chips at −10 dB and −0.47 chips at 2 dB (pulled towards the window centre), ≈ 0 above 14 dB.
2. **It is local.** $I(\tau)$ involves only s' at the true delay — the curvature of the *true* peak. It cannot know that other peaks exist (Figure 1).
3. **It ignores the prior.** Below about −8 dB the CRLB exceeds the prior variance $T_a^2/12$, i.e. it "allows" more error than knowing nothing — useless as a bound there.

Edge effect (small). For delays within 0.05 chip of a window edge, the constrained MLE achieves MSE/CRLB = 0.63 at 20 dB (half of its possible errors are cut off). This changes the averaged MSE by about 1 %.

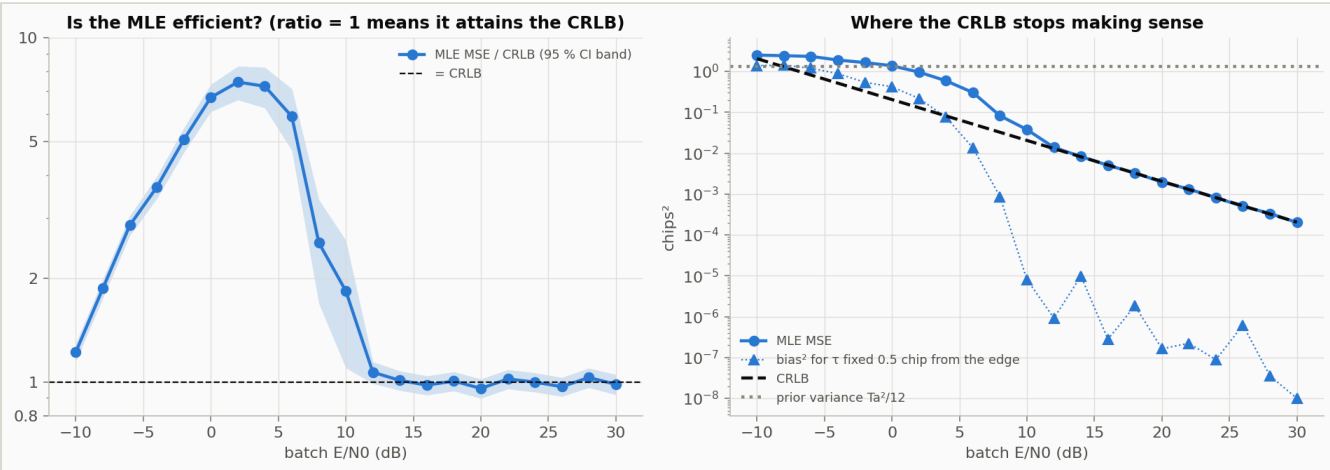


Figure 5. Left: MLE MSE/CRLB with 95 % band — 1 above ≈ 12–14 dB. Right: MSE, CRLB, prior variance and the conditional bias² for a delay near the edge.

6 The threshold effect

Above threshold the estimate always lands on the true correlation peak and errors are small and Gaussian. In the threshold region a small fraction P of trials are **gross errors** (|e| > 0.5 chip, outside the main lobe). A useful mental model (the idea behind the method of interval errors, MIE):

$$\mathrm{MSE} \approx (1 - P) \cdot \mathrm{CRLB} + P \cdot \sigma^2_{\mathrm{gross}},$$

where $\sigma^2_{\mathrm{gross}}$ is the mean-square size of a wrong-peak error (of order the prior width squared). Because $\sigma^2_{\mathrm{gross}} \gg \mathrm{CRLB}$, even $P = 1\%$ dominates: at 10 dB a 1 % rate of 2-chip errors adds 0.04 chips², twice the CRLB. The threshold is therefore a change in the **type** of error, not a gradual widening.

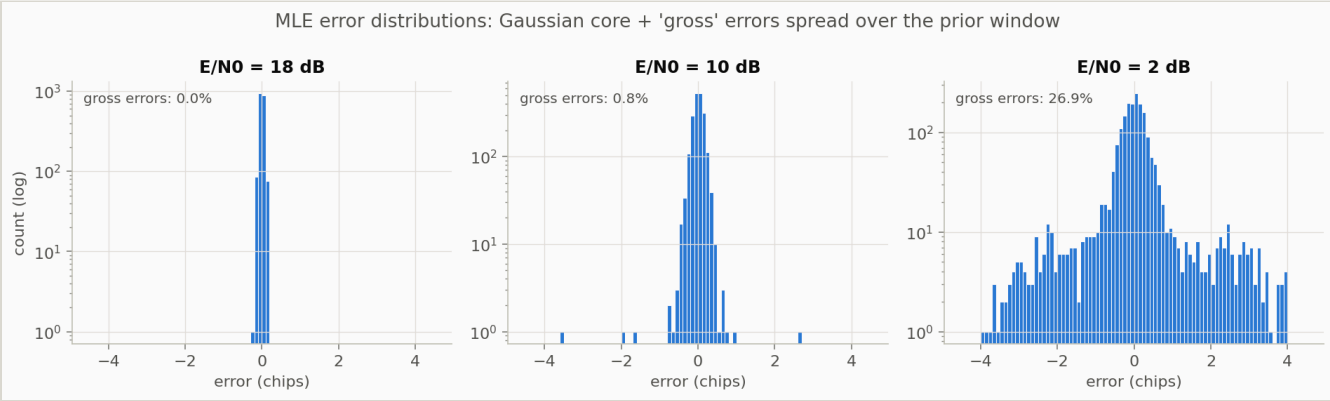


Figure 6. MLE error histograms (log count) at 18, 10 and 2 dB: a narrow Gaussian core, then a growing population of gross errors spread over the window.

Very low SNR. On pure noise the MLE's MSE is 3.09 chips² — larger than the prior variance 1.33 — because the arg-max of a noise process piles up at the window edges (about 20 % of estimates sit exactly on an edge). The MMSE estimator instead tends to the window centre and its MSE to $T_a^2/12$.

7 Sprint 4 — the Ziv–Zakai bound, derived for this problem

Notebook 04. **DONE** (derivation) · numerical evaluation also done (Sprint-5 material).

7.1 The idea

If an estimator could locate τ to within $h/2$, it could tell "$\tau = \varphi$" from "$\tau = \varphi + h$". Detection theory gives the best possible error rate for that test; therefore the estimator must make errors of size about $h/2$ at least that often. Doing this for **every** separation h — small and large — and adding up gives a lower bound on the MSE that sees both local and distant (wrong-peak) errors.

7.2 Derivation, step by step

Step 1 — MSE as a sum over error sizes. Apply Section 2.6 to $X = |e|$ and substitute $x = h/2$ ($2x\,dx = \frac{h}{2}dh$):

$$\mathrm{MSE} = \int_0^\infty \frac{h}{2}\, P\big(|e| \ge \tfrac{h}{2}\big)\, dh. \tag{1}$$

Step 2 — split by direction and shift. Condition on the true delay and substitute $\varphi \to \varphi + h$ in the second part:

$$P(|e| \ge \tfrac{h}{2}) = \int \Big[p(\varphi) P(e \ge \tfrac{h}{2} \mid \varphi) + p(\varphi + h) P(e \le -\tfrac{h}{2} \mid \varphi + h) \Big] d\varphi. \tag{2}$$

Step 3 — recognise a detector. Fix φ, h and consider $H_0 : \tau = \varphi$ vs $H_1 : \tau = \varphi + h$ with weights $p(\varphi), p(\varphi + h)$. Use the estimator to decide: *choose H_1 iff $\hat\tau \ge \varphi + h/2$*. Under H_0 it errs exactly when $e \ge h/2$; under H_1 exactly when $e < -h/2$. So the bracket in (2) is $\big(p(\varphi) + p(\varphi+h)\big)\times$ (error probability of this detector), which is at least that of the optimal detector:

$$P(|e| \ge \tfrac{h}{2}) \;\ge\; \int \big(p(\varphi) + p(\varphi + h)\big)\, P_{\min}(\varphi, \varphi + h)\, d\varphi. \tag{3}$$

Step 4 — combine and valley-fill. Substituting (3) into (1):

$$\mathrm{MSE} \;\ge\; \frac{1}{2}\int_0^\infty h\, \mathcal{V}\Big\{ \int \big(p(\varphi) + p(\varphi+h)\big) P_{\min}\, d\varphi \Big\}\, dh, \qquad \mathcal{V}\{g\}(h) = \max_{\eta \ge 0} g(h + \eta). \tag{4}$$

Valley filling is allowed because the left side of (3) can only decrease with h, so it also exceeds every later value of the right side (Bell et al., 1997). For our pilot it changes the bound by less than 0.01 %.

Step 5 — the optimal detector for our signals. H_0, H_1 are two known signals $s(\cdot - \varphi)$, $s(\cdot - \varphi - h)$ in white Gaussian noise, equally likely under the uniform prior. By Section 2.3, $P_{\min} = Q(\|\mathbf{s}_1 - \mathbf{s}_0\|/2\sigma)$, and $\|\mathbf{s}_1 - \mathbf{s}_0\|^2 = \|\mathbf{s}_0\|^2 + \|\mathbf{s}_1\|^2 - 2\langle \mathbf{s}_0, \mathbf{s}_1\rangle = 2\sum s^2(1 - \rho(h))$. With $\sum s^2/(2\sigma^2) = E/N_0$:

$$P_{\min}(h) = Q\left(\sqrt{\tfrac{E}{N_0}\big(1 - \rho(h)\big)} \right). \tag{5}$$

Two delays are hard to distinguish when the signal is weak *or* when the shifted signals look alike ($\rho(h)$ near 1: inside the main lobe or on a side-lobe). $P_{\min}$ does not depend on φ (verified to $2 \cdot 10^{-5}$).

Step 6 — the uniform prior. Both φ and $\varphi + h$ must lie in the window: a range of length $T_a - h$, each density $1/T_a$, so $\int (p(\varphi) + p(\varphi+h))d\varphi = 2(1 - h/T_a)$ for $0 \le h \le T_a$. The factors $\frac{1}{2}$ and 2 cancel:

RESULT — THE ZZB FOR OUR PROBLEM

$$\mathrm{ZZB} = \int_0^{T_a} h\, \mathcal{V}\left\{\left(1 - \frac{h}{T_a}\right) Q\left(\sqrt{\frac{E}{N_0}\,(1-\rho(h))}\right)\right\} dh.$$

7.3 The two limits — proofs

NO SIGNAL: ZZB → $T_A^2/12$

$E/N_0 \to 0$ gives $Q(0) = \frac{1}{2}$: $\int_0^{T_a} \frac{h}{2}(1 - \frac{h}{T_a})dh = \frac{T_a^2}{4} - \frac{T_a^2}{6} = \frac{T_a^2}{12}$, the prior variance. No estimator can do better without data, and "answer the window centre" achieves it. Unlike the CRLB, the bound stays finite. ■

STRONG SIGNAL: ZZB → CRLB

Only small h contribute (otherwise $Q \approx 0$). There $1 - \rho(h) \approx \frac{1}{2}\beta^2 h^2$ (Section 2.5) and $1 - h/T_a \approx 1$. With $a^2 = \frac{1}{2}(E/N_0)\beta^2$: $\mathrm{ZZB} \to \int_0^\infty h\, Q(ah)\, dh$. Integrate by parts with $Q'(x) = -\phi(x)$:

$$\int_0^\infty hQ(ah)\,dh = \left[\tfrac{h^2}{2}Q(ah)\right]_0^\infty + \frac{a}{2}\int_0^\infty h^2\phi(ah)\,dh = \frac{1}{2a^2}\int_0^\infty u^2\phi(u)\,du = \frac{1}{2a^2}\cdot\frac{1}{2} =$$

and $1/(4a^2) = 1/(2(E/N_0)\beta^2) = \mathrm{CRLB}$. ■

Numerically: the ZZB equals 1.3323 at zero SNR ($T_a^2/12 = 1.3333$) and ZZB/CRLB = 0.997 at 40 dB.

7.4 Intuition and how to explain it

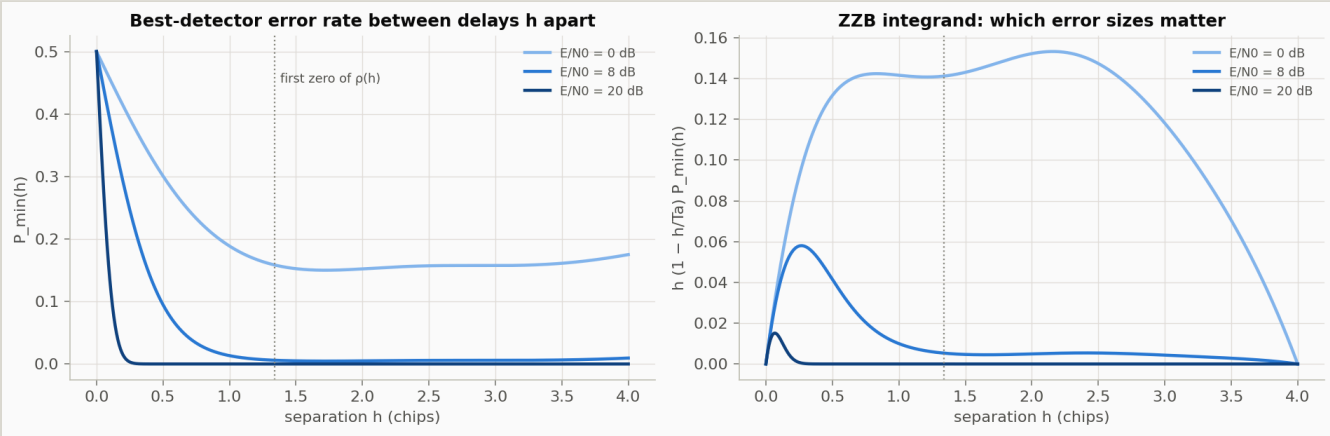


Figure 7. Left: $P_{\min}(h)$ at 0, 8 and 20 dB. Right: the integrand $h(1 - h/T_a)P_{\min}(h)$ — at 20 dB only tiny separations contribute (local, CRLB-like); at 0 dB separations far outside the main lobe contribute heavily (the gross errors).

SAY IT SIMPLY

"Suppose an estimator is usually within half a chip. Then I could use it to tell 'delay φ' from 'delay φ+1 chip'. Detection theory says how often *any* detector must confuse those two — that depends only on the SNR and how similar the two shifted signals are. So the estimator must make errors that big at least that often. Doing this for every separation and adding up gives a lower bound on the MSE. At high SNR only tiny separations get confused and I recover the CRLB; at low SNR even distant delays are confused and the bound rises — that's the threshold. With no signal, the bound is the prior variance."

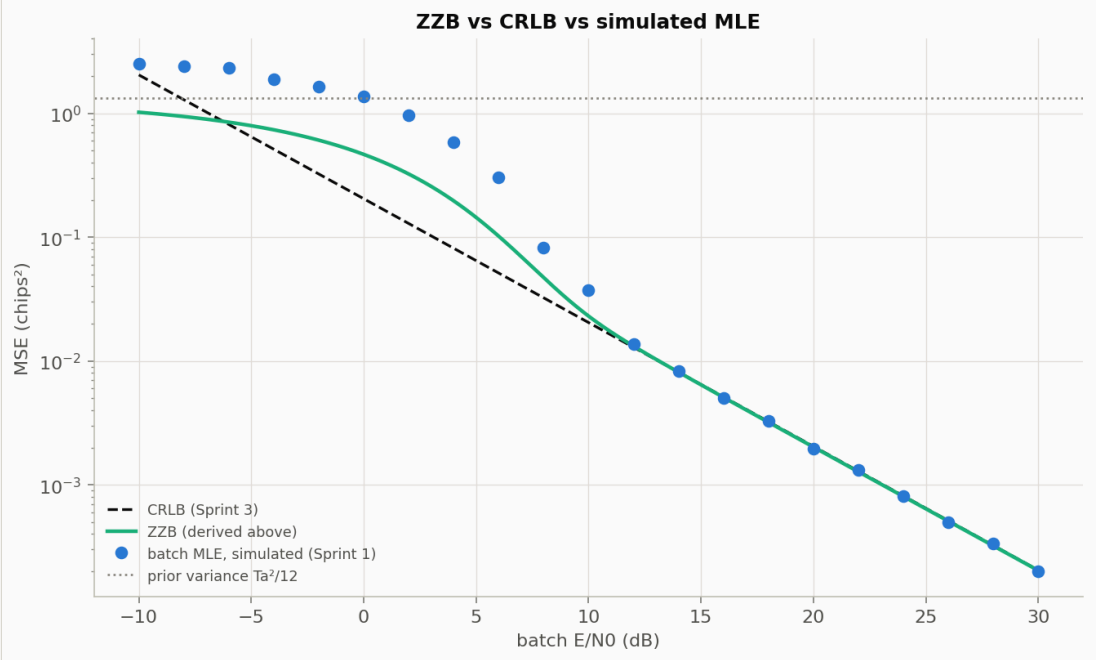


Figure 8. The ZZB reproduces both limits, bends away from the CRLB at ≈ 6 dB, and lies below the simulated MLE everywhere (within the 95 % Monte Carlo interval).

8 Sprint 2 — the early–late delay-locked loop

Notebook 02. DONE

8.1 The discriminator

With current estimate θ and error $\varepsilon = \theta - \tau$, correlate the data with replicas shifted by $\pm d/2$ ($d = 0.5$ chip):

$$E = \sum_n r[n]\, s(t_n - \theta + \tfrac{d}{2}), \qquad L = \sum_n r[n]\, s(t_n - \theta - \tfrac{d}{2}).$$

Without noise each is a sample of the autocorrelation $R(h) = \sum_n s(t_n)s(t_n - h)$: $E - L = R(\varepsilon - \frac{d}{2}) - R(\varepsilon + \frac{d}{2})$. R is symmetric with its peak at 0, so $E = L$ when $\varepsilon = 0$, and a Taylor expansion gives

$$E - L \approx -2R'(\tfrac{d}{2})\,\varepsilon = k\,\varepsilon, \qquad k > 0 \text{ (since } R'(d/2) < 0).$$

Dividing by k, the discriminator $D = (E - L)/k \approx \varepsilon$ measures the error. The whole curve $D(\varepsilon)$ is the **S-curve**.

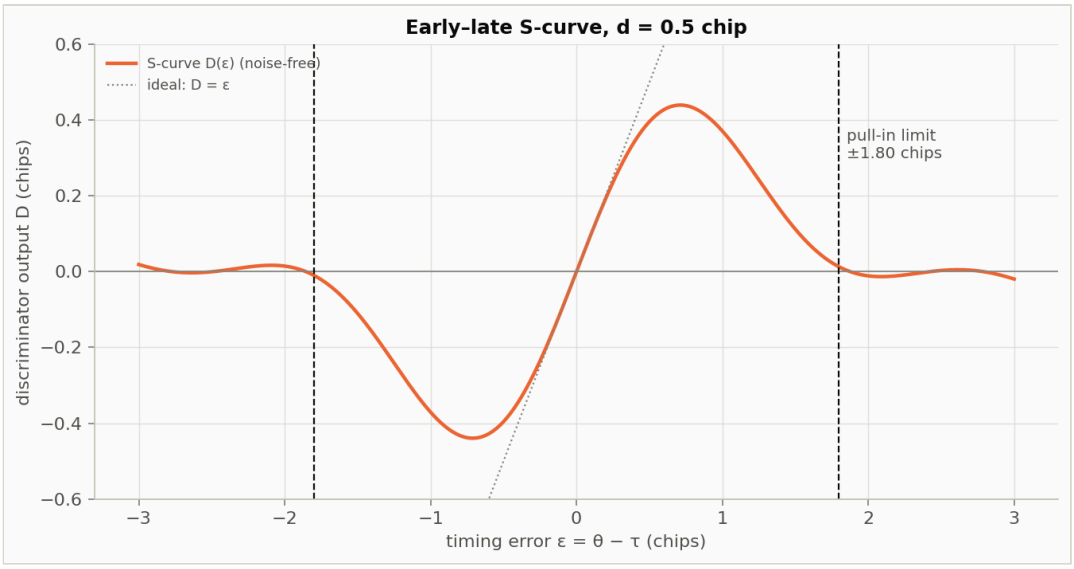


Figure 9. The S-curve: slope 1 at zero, peak 0.44 at ε = 0.70 chip, correct sign out to ±1.80 chips (pull-in range).

8.2 The loop and its noise averaging — derivation

Every $M = 4$ chips: $\theta_{j+1} = \theta_j - K\,D_j$. In the linear region $D_j = \varepsilon_j + n_j$, so

$$\varepsilon_{j+1} = (1 - K)\,\varepsilon_j - K\,n_j.$$

- **Stability:** errors decay iff $|1 - K| < 1$, i.e. $0 < K < 2$.
- **Noise averaging:** the error is the noise filtered by $h_k = K(1 - K)^k$. For independent n_j of variance σ_n^2, $\mathrm{var}(\varepsilon) = \sigma_n^2 \sum_k h_k^2 = \sigma_n^2 \frac{K^2}{1-(1-K)^2} = \sigma_n^2 \frac{K}{2-K}$.
- Averaging N independent measurements gives σ_n^2/N, so the loop is equivalent to averaging $N_{\mathrm{eff}} = (2 - K)/K$ updates; its noise bandwidth is $B_L T_I = \sum h_k^2 / (2(\sum h_k)^2) = K/(2(2 - K))$.

Design: $N_{\mathrm{eff}} = 64/4 = 16 \Rightarrow K = 2/17 \approx 0.118 \Rightarrow 1/(2B_L) = 64$ chips: the locked loop averages as much signal as the 64-chip batch. The loop runs 160 updates = 640 chips in total.

8.3 Implementation details that were necessary (each found by testing)

Detail	What we do	What went wrong without it
Acquisition	coarse search (0.5-chip grid) over the prior on the first 64 chips, then track	the loop only pulls in within ±1.8 chips; the prior is ±2
Block replicas	block j's replica contains only its 4 chips	full replica over a hard-cut window: bias ≈ 33 % of the CRLB at 30 dB
Offset calibration b_j	subtract each block's noise-free discriminator output at zero error	neighbour-chip overlap gives a fixed offset: MSE ≈ 2× CRLB at 30 dB. b_j depends only on the known pilot (identical results with deliberately wrong timing)
Constant gain $\bar{k}$	normalise by the average slope	per-block slopes vary 0.29–1.7× the average; dividing by small ones amplifies noise
Windows follow θ	integration windows placed by the loop's own timing	fixed windows distorted low-SNR behaviour
Same pilot, repeated	the 64-chip pilot repeated like a GNSS code	a different pilot made the DLL appear to beat the CRLB

8.4 Locked accuracy and its theory

Linearised, the final error is a weighted sum of the noise samples, $\varepsilon = \langle \mathbf{w}, G \rangle$, with $G = \sum_j c_j g_j$ (g_j = early-minus-late replica of block j, $c_j = (K/\bar{k}) \prod_{i>j}(1 - K k_i/\bar{k})$). Hence exactly $\mathrm{var}(\varepsilon) = \sigma^2 \|G\|^2$. Measured: DLL MSE/CRLB = 1.088, 1.083, 1.039 (±0.024) at 15, 20, 25 dB; theory 1.071.

> **BE PRECISE ABOUT THIS NUMBER**
> "DLL ≈ CRLB" holds **because we chose K**. A narrower loop averages more of its 640 chips and goes *below* the 64-chip CRLB — not a violation, because that CRLB is not a bound for an estimator using 640 chips. The 7 % excess comes from early–late being a finite-difference approximation of the derivative an optimal estimator would use.

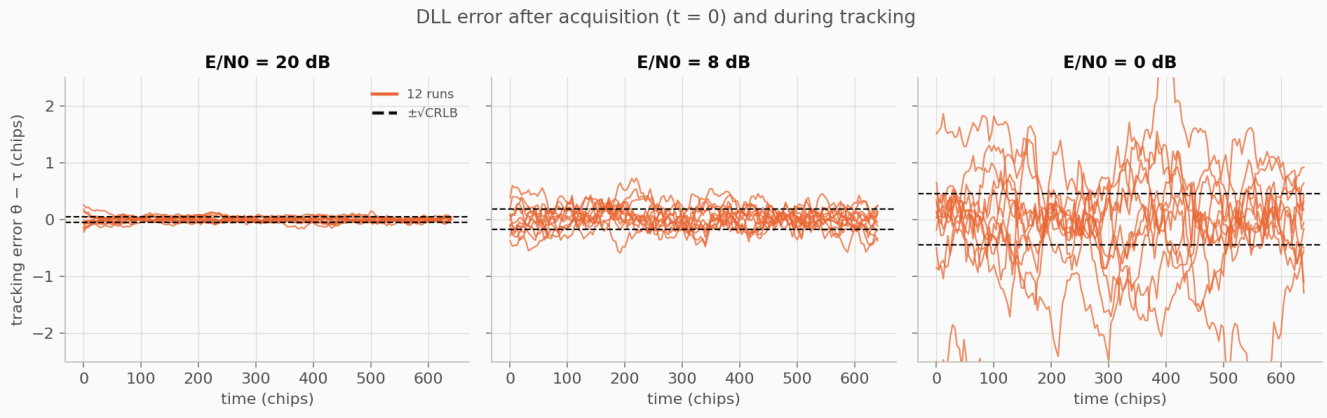


Figure 10. DLL error over time, 12 runs at 20, 8 and 0 dB: tight lock, noisy lock, and loss of lock.

9 Experimental methodology and validation

9.1 Monte Carlo

For each SNR we draw N independent trials (random delay from the prior, fresh noise, one estimate) and report $\widehat{\mathrm{MSE}} = \frac{1}{N}\sum_i e_i^2$. Its standard error is $\mathrm{sd}(e^2)/\sqrt{N}$. Near the threshold e^2 is heavy-tailed (rare large errors), so the MSE is uncertain exactly where we want to locate the threshold — the reason for many trials: 2000 per point (notebooks 01, 05), 20,000 (notebook 06, MLE/MMSE), 10,000 (notebook 07), 4000 (DLL).

9.2 Threshold definitions (fixed before looking at results)

- **MSE criterion:** the highest SNR at which MSE/CRLB first exceeds 2 (3 dB), scanning down from high SNR.
- **Gross-error criterion:** the highest SNR at which $P(|e| > 0.5 \text{ chip})$ first exceeds 1 %. One outlier moves this by only $1/N$, whereas it can multiply the MSE.

Thresholds are interpolated linearly in (dB, log y) between grid points.

9.3 Confidence intervals by bootstrap

Resample the N trials at each SNR with replacement, recompute the threshold, repeat 200–300 times, report the 2.5–97.5 % percentiles. Differences (e.g. DLL – MLE) are computed sample by sample from the bootstrap draws so the interval includes both uncertainties. MLE and MMSE share the same noise (paired), which makes their difference precise.

9.4 How we know the code is right

Check	Result
Analytic checks in every notebook (Nyquist property, SNR calibration, noise-free estimators, ZZB limits, S-curve slope)	all pass
Fast regression tests (`tests/test_core.py` , 7 tests against analytic facts)	pass in ≈ 35 s
Independent from-scratch re-implementation (Oct 7)	CRLB agrees to 0.02 %, ZZB to 0.12 %, MLE MSE within Monte Carlo error
Adversarial code audit (Oct 7)	no estimator uses the true delay; DLL calibration independent of timing phase
Literature check of every formula (Oct 7)	ZZB form, $P_{\min}$, limits, CRLB, loop formulas all correct

All random seeds are fixed, so every number in this guide reproduces exactly.

10 Results

10.1 Sprint 1 — the MLE follows the CRLB, then breaks away

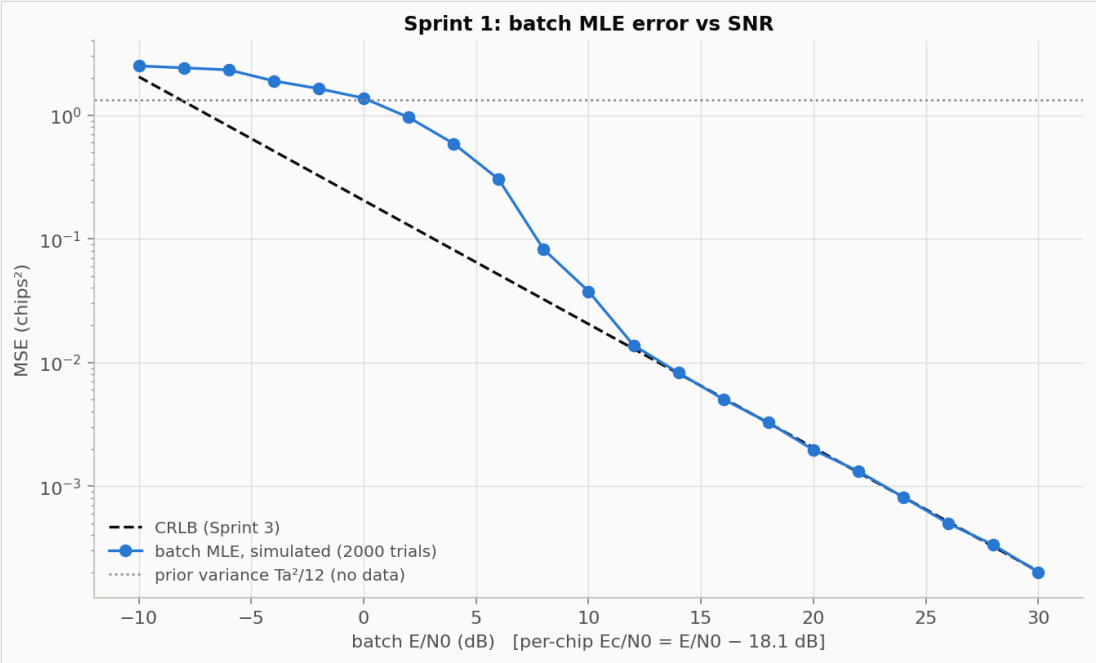


Figure 11. Batch MLE MSE (2000 trials/point), CRLB and prior variance.

Regime	Evidence	Meaning
High SNR (≥ 12–14 dB)	MSE/CRLB = 0.997 on average over 14–30 dB (points 0.96–1.03)	the MLE is efficient: it attains the CRLB
Threshold (≈ 6–12 dB)	gross errors 0.8 % at 10 dB, 2.1 % at 8 dB, 8.7 % at 6 dB	wrong-peak errors take over
Low SNR (≤ 0 dB)	MSE above the prior variance; 3.09 chips² on pure noise	no useful information; edge pile-up

10.2 Sprint 2 — the DLL works and has a limited pull-in range

Slope of the S-curve at zero 1.000; pull-in range ±1.80 chips; noise-free final error ≈ 10^{-7} chips; locked MSE 1.04–1.09 × CRLB (by design of K); at 0 dB some runs lose lock (Figure 10).

10.3 Sprint 3 — the CRLB is right where it should be and wrong where it must be

Derived and implemented CRLB agree with a frequency-domain computation to 0.05 %; MLE attains it above threshold; three identified reasons it fails below (Section 5.4); pilot dependence factor 2.4.

10.4 Sprint 4 — the ZZB is a valid, informative bound

Both limits reproduced (1.3323 vs 1.3333; 0.997 × CRLB at 40 dB); the bound lies below the simulated MLE at every SNR within the 95 % interval; it departs from the CRLB at 5.95 dB.

10.5 Sprint 5 — four curves on one plot

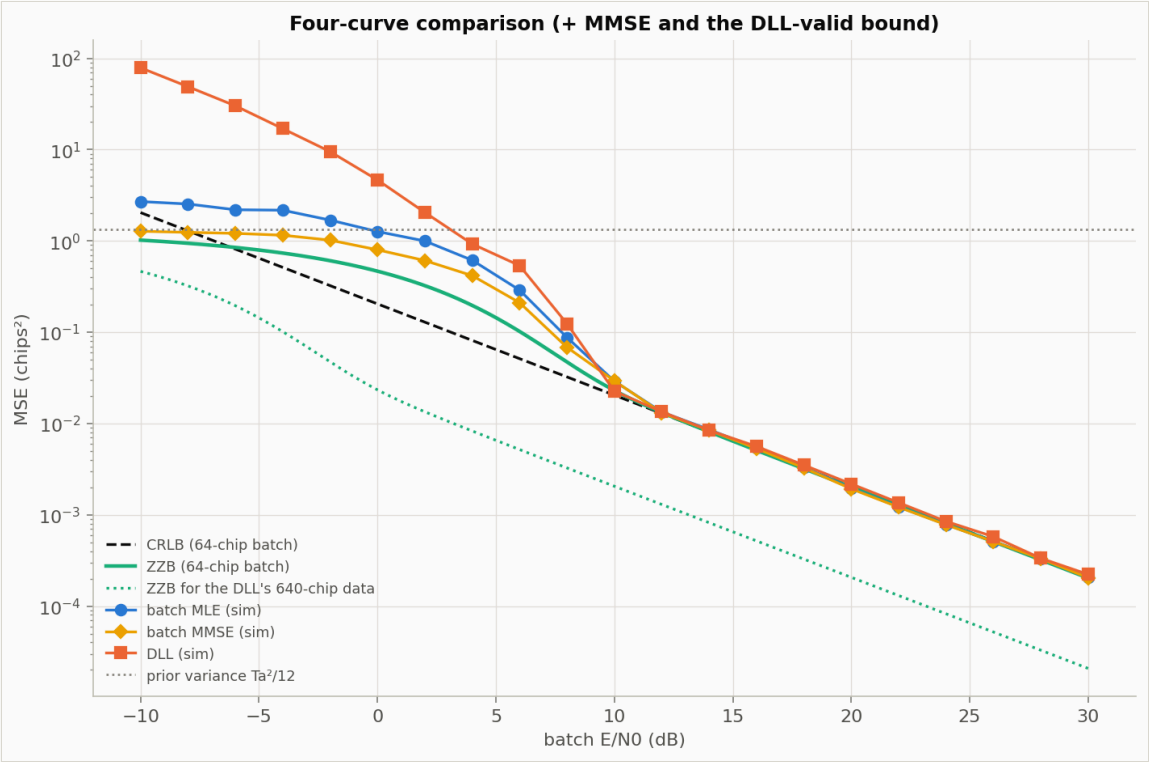


Figure 12. CRLB, ZZB, batch MLE, batch MMSE and DLL (2000 / 1000 trials per point), plus the ZZB for the DLL's own 640-chip data (dotted) — the only one of the bounds that is valid for the DLL.

- Above ≈ 12 dB everything sits on the 64-chip CRLB (DLL slightly above, by design).
- At low SNR the MMSE flattens at the prior variance as the ZZB predicts (−10 dB: MMSE 1.28, ZZB 1.02, MLE 2.70).
- Below threshold the DLL is far worse than both batch estimators, and it stays above the ZZB for its own data.

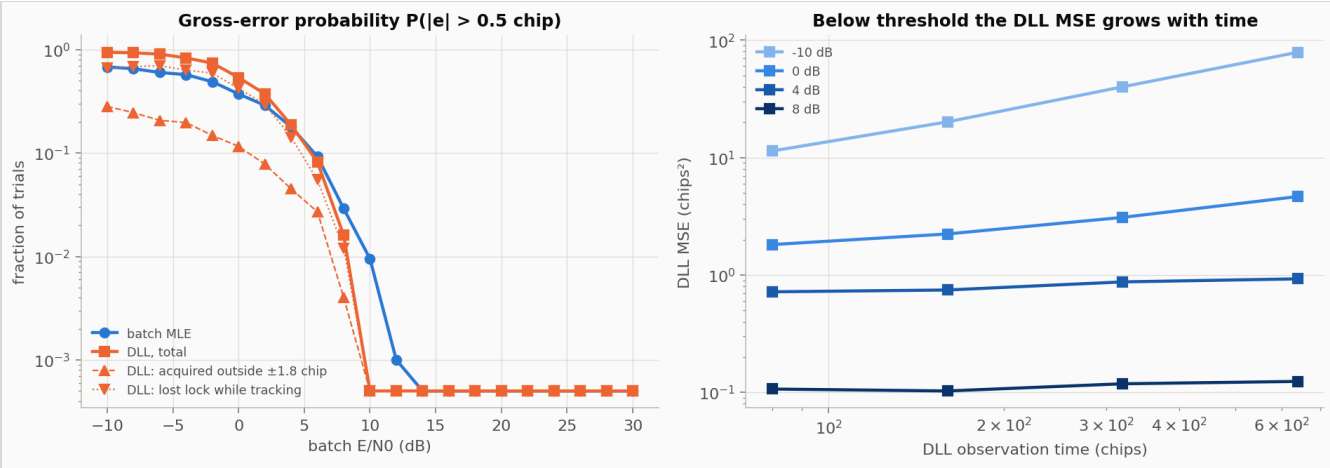


Figure 13. Left: gross-error probability; DLL failures split into "acquired outside the pull-in range" and "lost lock while tracking" (the two add up to the total). Right: below threshold the DLL's MSE grows with observation time.

At 0 dB, 54 % of DLL runs end with a gross error: 12 % acquired outside the pull-in range, 42 % lost lock while tracking. Loss of lock is the dominant DLL failure — a mechanism the batch MLE does not have.

10.6 Sprint 6 — the threshold, located precisely

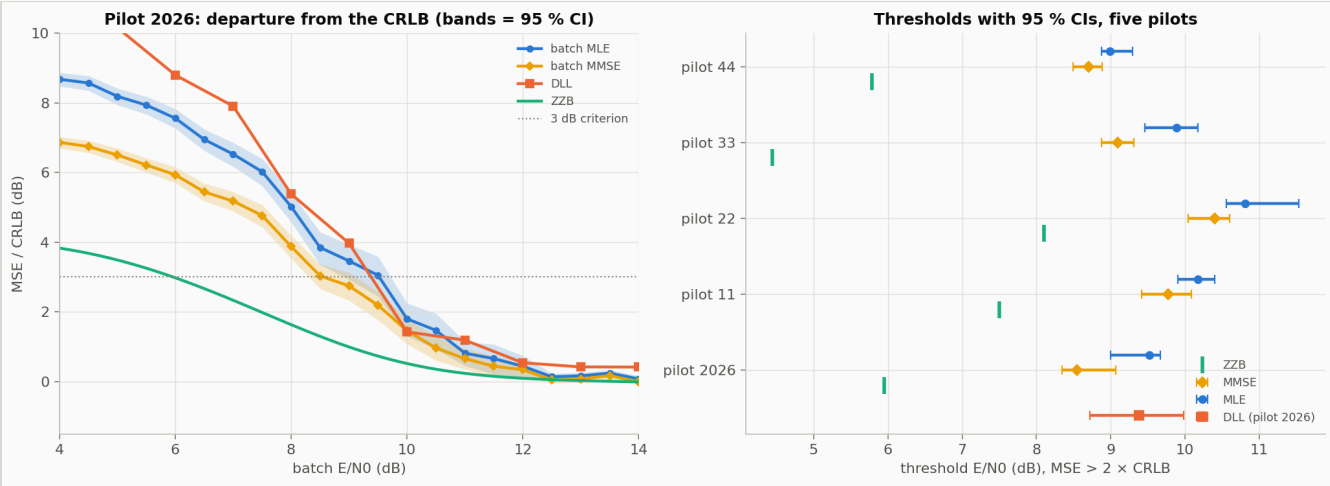


Figure 14. Left: departure from the CRLB with 95 % bands (pilot 2026; 20,000 trials/point for MLE/MMSE, 4000 for the DLL). Right: thresholds with 95 % CIs for five random pilots.

Pilot 2026, MSE criterion	Threshold	95 % CI
ZZB (bound)	5.95 dB	—
Batch MMSE (best possible estimator)	8.54 dB	[8.35, 9.07]
Batch MLE	9.52 dB	[9.00, 9.66]
DLL	9.38 dB	[8.72, 9.98]

The gap to the bound splits in two: bound looseness MMSE − ZZB = 2.59 dB, and MLE sub-optimality MLE − MMSE = 0.85 dB (CI 0.25–1.24, significant). About three-quarters of the gap is the bound being optimistic — built from *pairwise* tests while an estimator must reject *all* competing delays.

Five pilots: MLE − ZZB ranges 2.7–5.4 dB (mean 3.5); MMSE − ZZB 2.3–4.7 dB (mean 2.9); the MMSE is ahead of the MLE for every pilot. The ZZB threshold moves a lot (4.4–8.1 dB) and nearly follows each pilot's β^2, while the MLE threshold moves less (9.0–10.8 dB): because thresholds are defined relative to each pilot's own CRLB, a pilot with larger β^2 leaves its (lower) CRLB sooner.

DLL vs MLE depends on the criterion:

- MSE criterion: DLL − MLE = −0.12 dB, CI [−0.87, 0.59] → **not significant**.
- Gross-error criterion: DLL 8.81 dB vs MLE 9.49 dB → the DLL is **0.68 dB more robust** (CI 0.40–0.99): it observes 640 chips and recovers from wrong acquisitions within its pull-in range.
- Far below threshold the DLL is much worse (loss of lock, unbounded drift).

A subtlety: the MMSE has a lower MSE threshold but a slightly higher gross-error threshold than the MLE (9.63 vs 9.49 dB): by hedging between peaks, it makes more moderate errors just above 0.5 chip and fewer very large ones.

10.7 Supplement — the prior width moves the threshold

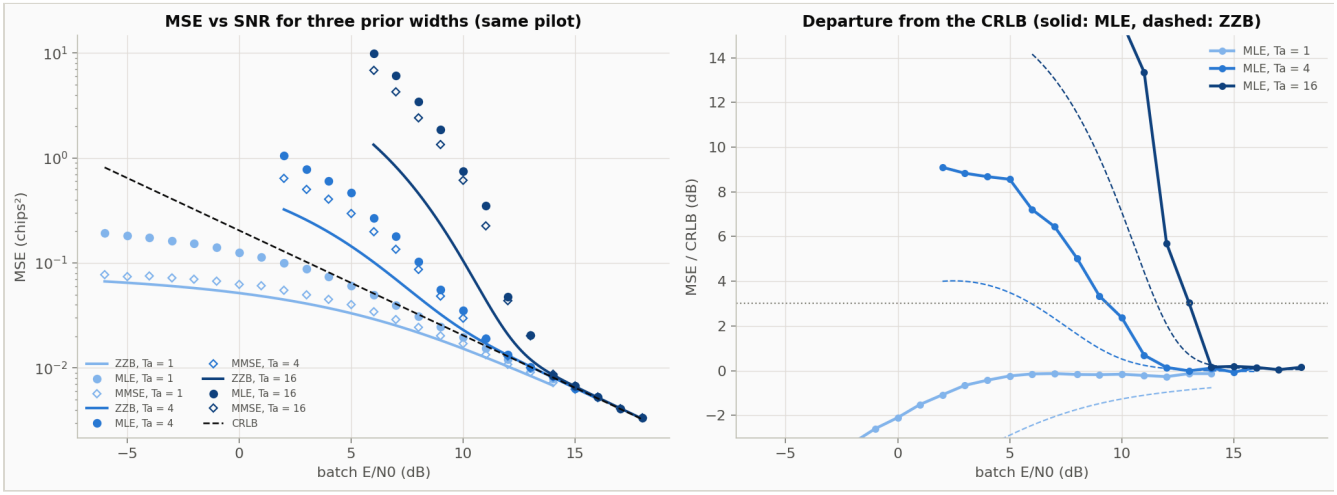


Figure 15. Same pilot, prior widths 1, 4 and 16 chips (10,000 trials/point).

T_a	ZZB	MMSE (95 % CI)	MLE (95 % CI)	MLE − ZZB
1 chip	none	masked	masked	—
4 chips	5.95 dB	8.84 [8.58, 9.14]	9.34 [8.81, 10.08]	3.4 dB
16 chips	11.56 dB	12.99 [11.97, 13.37]	13.01 [11.95, 13.41]	1.5 dB

- A wider window has more places for a winning noise peak, so the threshold moves up ≈ 3.7 dB from $T_a = 4$ to 16.
- With $T_a = 1$ there is no distant peak: the threshold is **masked** (the MSE never exceeds 2×CRLB).
- The ZZB is relatively tighter for the wide prior (gap 3.4 → 1.5 dB) — an observation we have not yet explained.

11 Decision log and the mistakes we fixed

Being able to say *why* each choice was made, and what went wrong before, is what convinces an examiner that the work is understood.

When	Decision / correction	Why
Sprint 1	m-sequence script replaced by the raised-cosine, random-delay model	ZZB needs a prior; exact delay; analytic spectrum
Oct 6	Prior width 1 → 4 chips	with 1 chip the MLE never left the CRLB (threshold masked)
Oct 6	DLL given the MLE's pilot, repeated periodically	with a different pilot the DLL appeared to beat the CRLB (the CRLB is pilot-specific)
Oct 6	Block replicas, offset calibration, constant gain	removed biases of 33 % and 100 % of the CRLB and a noise blow-up
Oct 7	Error bars and two threshold criteria	validation showed 2000 trials on a 2-dB grid cannot support "the DLL is 1.2 dB worse"
Oct 7	Pull-in range corrected 0.75 → 1.8 chips; failure classes made to partition the total	measured S-curve; overlapping categories were wrong
Oct 7	"DLL ≈ CRLB is a bound" withdrawn	the match holds by design of K; the 64-chip CRLB does not bound a 640-chip estimator
Oct 7	ZZB for the DLL's own data added	only valid bound for the DLL
Oct 7	DLL windows follow its own timing	fixed windows distorted low-SNR behaviour
Oct 9	Pulse tapered over its last chip	hard cut made the ZZB miss the CRLB limit by ~1 %
Oct 9–12	"MLE biased (bias² curve)" replaced by conditional bias; "gap mostly MLE sub-optimality" corrected	averaged bias cancels by symmetry; MMSE only ≈ 0.85 dB better
Nov 5	"DLL not different from MLE" refined	true for MSE, false for gross errors (notebook 06)
Nov 5	Novelty claim narrowed	batch ZZB for GNSS code tracking already exists (Survey 04)

12 What the four surveys taught us

Survey	Method	Lesson
01 Independent validation (Oct 7)	literature check, from-scratch re-implementation, adversarial audit	core theory and code correct; several *claims* wrong → fixes and error bars
02 Literature survey (Oct 13)	annotated bibliography, gap searches	everything up to the batch threshold is a reproduction; candidate gap: a bound for DLL loss of lock
03 Deep-research novelty check (Oct 13)	multi-agent search, 25 claims × 3 verifiers	optimal-MTLL theory exists for *phase* tracking → claim must be about code delay and the bound↔MTLL link
04 Interdisciplinary & second novelty check (Nov 5)	targeted searches	batch ZZB for GNSS code tracking already exists (Emmanuele 2012, Liu 2016); natural bridge = state estimation + integrity; robotics/VLM links would be forced

> **CUMULATIVE POSITION**
> **Reproduction:** MLE, CRLB, ZZB for delay, MLE threshold prediction (Mallat et al. 2014), batch ZZB for GNSS code tracking, linear DLL theory, DLL MTLL. **Our own observations:** ZZB looseness measured against the MMSE (2.6 dB + 0.85 dB), its pilot dependence, the DLL failure decomposition, the criterion-dependent DLL-vs-MLE comparison, the prior-width trend of the gap. **Appears open (medium-low confidence):** bounds with tracking dynamics linked to loss of lock for PN code delay. **Rule:** always "we found no prior work that…", never "first".

13 Limitations

- Synthetic data only (needed for ground truth and bound comparison).
- Constant delay (no Doppler, so a first-order loop suffices); coherent tracking with known amplitude and phase.
- Known pilot (no unknown data bits); no multipath.
- One DLL design (d = 0.5, 4-chip updates, one loop gain).
- The batch ZZB is not a bound for the DLL; the 640-chip ZZB is valid but loose.
- The pilot-to-pilot spread and the prior-width trend of the gap are measured but not explained theoretically.
- Novelty of the proposed extension is uncertain (absence-of-evidence argument).

14 Proposal: where does a delay *tracker* lose information?

PROPOSED — nothing in this section is implemented. Full text: `docs/evaluation/research_proposal.md`.

14.1 Directions considered

Direction	Novelty	Feasible now	Verdict
D1 Dynamic decomposition of the tracking threshold (bound / estimator / loop)	medium	yes (core)	**recommended**
D2 Loss-of-lock risk prediction (safety-critical autonomy)	medium	yes (reuses D1)	part of D1 (H4)
D3 Adaptive loop bandwidth via control/RL	low (LBCA, fuzzy, adaptive KF, RL proof of concept exist)	partly	not pursued
D4 Pilot/code selection by threshold	low–medium (ZZB waveform design exists)	yes (≈1 week)	small side project
D5 Robot localisation with UWB	low; physics mismatch (NLOS ≠ thermal threshold)	risky	forced — not pursued

14.2 Research question and intuition

Our static study split the gap to the bound into *bound looseness* (ZZB → MMSE, 2.6 dB) and *estimator sub-optimality* (MMSE → MLE, 0.85 dB). The project is about *tracking*, where two new things happen: **memory** (the prior at each step is the previous posterior, which narrows the search window and should lower the threshold) and **feedback** (the DLL commits to one estimate and can be pushed outside its pull-in range — loss of lock). **Question:** for a moving delay, how much of a DLL's low-SNR failure is fundamental, how much is sub-optimal estimation, and how much is loss of lock — and can loss of lock be predicted quantitatively?

14.3 Model

$$d_k = d_{k-1} + w_k, \quad w_k \sim \mathcal{N}(0, q); \qquad r_k[n] = s_k(t_n - d_k) + v_k[n], \quad v_k[n] \sim \mathcal{N}(0, \sigma^2),$$

one update k per block of M chips. Likelihood (Section 4.1): $\ln p(\mathbf{r}_k \mid d) = J_k(d)/\sigma^2 + \text{const}$.

14.4 The optimal tracker — an exact grid Bayes filter

$$p(d_k \mid \mathbf{r}_{1:k}) \;\propto\; \underbrace{e^{J_k(d_k)/\sigma^2}}_{\text{update}} \underbrace{\int \mathcal{N}(d_k; d_{k-1}, q)\, p(d_{k-1} \mid \mathbf{r}_{1:k-1})\, dd_{k-1}}_{\text{prediction} \,=\, \text{Gaussian convolution}}.$$

Because the state is one-dimensional, the recursion can be computed essentially exactly on a fine grid (point-mass filter, Bucy & Senne 1971) — so the "optimal" baseline carries no Monte Carlo approximation error. Its posterior mean is the MMSE tracker.

14.5 The posterior CRB for this model — derivation

Tichavský, Muravchik & Nehorai (1998) give $J_k = D_{22} - D_{21}(J_{k-1} + D_{11})^{-1} D_{12}$ with $D_{11} = \mathbb{E}[-\partial^2_{d_{k-1}} \ln p(d_k \mid d_{k-1})] = 1/q$, $D_{12} = D_{21} = -1/q$, $D_{22} = 1/q + J_D$, where $J_D = \sum_n s_k'(t_n)^2/\sigma^2$ is the measurement information (Section 5.1). Substituting:

$$J_k = \frac{1}{q} + J_D - \frac{1/q^2}{J_{k-1} + 1/q} = J_D + \frac{J_{k-1}}{q\,J_{k-1} + 1} = J_D + \frac{1}{q + 1/J_{k-1}}.$$

With $R = 1/J_D$ and $P = 1/J$, the fixed point satisfies $P^2 + qP - qR = 0$, so

$$P_\infty = \tfrac{1}{2}\Big(-q + \sqrt{q^2 + 4qR}\Big), \qquad K_\infty = \frac{P_\infty + q}{P_\infty + q + R},$$

exactly the steady-state scalar Kalman filter. **Link to our DLL:** $\theta_{k+1} = \theta_k - K D_k$ with $D_k \approx \varepsilon_k + n_k$ is a fixed-gain Kalman filter in the linear regime; choosing $K = K_\infty$ makes it optimal there. This replaces our Sprint-2 gain choice ("match 64 chips") with a principled one.

> **CHECK OF THE FIXED POINT**
> Kalman: prediction $P^- = P + q$, update $P = P^- R/(P^- + R)$. At steady state $P(P + q + R) = (P + q)R \Rightarrow P^2 + qP - qR = 0$. ∎

14.6 Loss of lock as a Markov chain — our own computation

Using the measured S-curve $D(\varepsilon)$ (Section 8.1) and discriminator noise of variance σ_D^2, the error evolves as

$$\varepsilon_{k+1} = \varepsilon_k - K\,D(\varepsilon_k) - K\,n_k + w_{k+1},$$

a one-dimensional Markov chain. Discretise ε on a grid inside the lock region $|\varepsilon| < R_{\text{pull}} \approx 1.8$ chips. The mean exit time $m(\varepsilon)$ satisfies the first-step equations

$$m(\varepsilon) = 1 + \sum_{\varepsilon'} P(\varepsilon \to \varepsilon')\, m(\varepsilon'), \qquad m = 0 \text{ outside the lock region},$$

i.e. one linear solve $(I - P_{\text{in}})\mathbf{m} = \mathbf{1}$; the mean time to lose lock is $m(0)$. Published MTLL results are mostly continuous-time, second-order loops at small-noise asymptotics; this discrete first-order version uses our measured S-curve directly and can be checked against simulation.

14.7 Risk score (applied part)

From the optimal tracker's posterior, the probability that the DLL is already outside its pull-in range: $\rho_k = P\big(|d_k - \hat\theta_k^{\text{DLL}}| > R_{\text{pull}} \mid \mathbf{r}_{1:k}\big)$. Compared with a classical correlator-power lock detector by ROC curve and detection lead time — relevant to integrity monitoring in autonomous navigation.

14.8 Hypotheses — each can fail

	Hypothesis	Test	If it fails
H1	Above threshold, DLL with $K = K_\infty$ and the grid tracker both attain the PCRB	RMSE vs E_c/N_0	implementation error or non-linear jitter — diagnose
H2	The DLL leaves the PCRB earlier than the grid tracker; conditioning on lock removes most of the excess	lock-conditioned RMSE	excess is jitter inflation inside lock — quantify it with the Markov chain
H3	Markov-chain MTLL matches simulation; the DLL threshold is where MTLL ≈ observation horizon	MTLL vs E_c/N_0, K	refine the discrete approximation
H4	The posterior risk score predicts loss of lock earlier than a classical detector	ROC/AUC, lead time	report where each detector wins

14.9 Experiments, baselines, metrics

- **E1** RMSE vs E_c/N_0: PCRB, grid tracker, linearised Kalman, DLL(K_∞) at 2–3 values of q, bootstrap CIs.
- **E2** DLL RMSE conditioned on staying in lock.
- **E3** MTLL: Markov chain vs simulation over E_c/N_0 and K.
- **E4** Loss-of-lock prediction: risk score vs correlator-power detector.
- **Ablations:** early–late spacing d; five pilots.
- **Stretch:** a sequential ZZB (Bell et al. construction on the one-step predictive prior) or the marginal WWB (Fritsche et al. 2015) — the most novel and most uncertain part.

14.10 Roadmap (≈ 3 weeks; adjust to the real calendar)

Week	Work	Deliverable	Priority
1	random-walk model, PCRB recursion, grid tracker, DLL(K_∞), E1; tests	notebook 08	P0
2	E2, Markov-chain MTLL (E3)	notebook 09	P0
2	risk score (E4), ablations	figures, table	P1
3	stretch bound; report and final presentation	report, slides	P0 / P2

14.11 How it differs from existing work

Existing	What it does	What D1 adds
Emmanuele et al. 2012; Liu et al. 2016	ZZB for one integration period of GNSS signals	tracking dynamics; decomposition bound / estimator / loop
Holmes & Biederman 1978; Hasson & Bobrovsky 1995; Groh et al. 2012	MTLL of given loops	link MTLL to the MSE threshold and to the optimal tracker
Tichavský 1998; Rapoport & Oshman 2007; Fritsche 2015	generic recursive bounds	applied to PN code tracking, against an exact optimal tracker
Fischler & Schuss 2007	optimal-MTLL filtering for phase	code delay (non-periodic, code-dependent side-lobes)

15 Question bank with follow-ups

Cover the answer, say it aloud, check. **F:** = likely follow-up.

A. Concepts

A1. What is your project in one sentence?

We estimate the delay of a known BPSK signal with a batch MLE and a DLL and compare them with the CRLB and the ZZB, to understand the low-SNR threshold where errors suddenly become large.

A2. What exactly is the threshold effect?

A change in the type of error: above threshold the estimate sits on the true correlation peak; below it, noise sometimes makes a wrong peak win and the estimate jumps chips away (Figure 1, Section 6).

F: Is it a sharp transition? — No, a region a few dB wide; hence fixed criteria and confidence intervals.

A3. Why plot MSE rather than variance?

Below threshold the estimators are biased for a given delay, and the ZZB bounds MSE. MSE = variance + bias² (Section 2.1).

A4. Why is the CRLB not valid at low SNR?

It assumes unbiasedness, it is local (curvature of the true peak only), and it ignores the prior — below −8 dB it exceeds the prior variance.

F: So is the CRLB wrong? — No; it is a correct bound for a class our estimators leave at low SNR. It is not informative there.

A5. Can an estimator beat the CRLB?

Yes: a biased one; a constrained one near the window edge (MSE/CRLB = 0.63 there); one using more data than the CRLB assumes (a narrow-band DLL uses 640 chips).

A6. Why does the ZZB need a prior? Is a uniform prior realistic?

It bounds the MSE averaged over the prior — that is why it stays finite at zero SNR ($T_a^2/12$). Uniform models "the delay is somewhere in this window" after coarse acquisition.

F: How does the result depend on T_a? — T_a = 1: threshold masked; 4 → 16: threshold rises ≈ 3.7 dB and the ZZB gap shrinks 3.4 → 1.5 dB (Section 10.7).

B. Model and implementation

B1. Why a raised-cosine pulse?

CRLB = $1/(2(E/N_0)\beta^2)$; rectangular chips have $\beta^2 = \infty \Rightarrow$ CRLB = 0. Real receivers band-limit.

B2. Why 4 samples per chip?

The spectrum ends at 0.75 cycles/chip < Nyquist (2); samples hold all information and sums equal integrals.

B3. Why did you taper the pulse?

A hard cut left a small slope jump; samples on it broke the identity $\rho''(0) = -\beta^2$, so the ZZB missed the CRLB limit by ~1 %. The taper only touches 5 < |t| ≤ 6, where integer points are zeros anyway.

B4. Why does the MLE need no SNR but the MMSE does?

σ^2 scales the whole log-likelihood (arg-max unchanged); the posterior mean weights by e^{J/σ^2}, which depends on it.

B5. Why did the model change after the first Sprint-1 script?

ZZB needs a random delay with a prior; exact analytic delay; analytic spectrum. Same qualitative result.

C. Derivations (expect the board)

C1. Derive the MLE.

Gaussian likelihood → expand the square → drop $\sum r^2$ → maximise $J(\theta)$ (Section 4.1).

C2. Derive the CRLB, including the inequality itself.

Score → $\mathbb{E}[(\hat{\theta} - \theta)S] = 1$ → Cauchy–Schwarz (Section 2.4); then $I = \frac{1}{\sigma^2}\sum s'^2$ (Section 5.1).

F: Why is β^2 the Gabor bandwidth? — Parseval: differentiation multiplies the spectrum by $j2\pi f$.

C3. Prove $\mathbb{E}[X^2] = \int_0^\infty 2xP(X \geq x)dx$.

$X^2 = \int_0^\infty 2x\mathbf{1}\{X \geq x\}dx$; take expectations, swap by Tonelli (Section 2.6).

C4. Why is any estimator's tail probability bounded by $P_{\min}$?

Use it as a detector (decide H_1 iff $\hat{\tau} \geq \varphi + h/2$); its errors are exactly the tail events; no detector beats the optimal one (Section 7.2, step 3).

C5. Why is $P_{\min} = Q(\sqrt{(E/N_0)(1-\rho(h))})$?

Equal priors, known signals in white noise: $Q(\|\mathbf{s}_1 - \mathbf{s}_0\|/2\sigma)$ (Section 2.3) and $\|\mathbf{s}_1 - \mathbf{s}_0\|^2 = 2\sum s^2(1-\rho)$.

C6. Show ZZB → CRLB at high SNR.

$1 - \rho \approx \frac{1}{2}\beta^2 h^2$; $\int_0^\infty hQ(ah)dh = 1/(4a^2)$ by parts; $a^2 = \frac{1}{2}(E/N_0)\beta^2$ (Section 7.3).

F: Why $\rho''(0) = -\beta^2$? — summation by parts: $\sum s\, s'' = -\sum s'^2$ (Section 2.5).

C7. Show ZZB → $T_a^2/12$ with no signal.

$Q(0) = \frac{1}{2}$; $\int_0^{T_a} \frac{h}{2}(1 - \frac{h}{T_a})dh = T_a^2/12$.

C8. What is valley filling and does it matter?

Replace the integrand by its non-increasing envelope (valid because the tail probability decreases in h). Here < 0.01 %.

C9. Derive the DLL's noise averaging.

$\varepsilon_{j+1} = (1-K)\varepsilon_j - Kn_j \to h_k = K(1-K)^k \to \sum h_k^2 = K/(2-K) \to N_{\text{eff}} = (2-K)/K$ (Section 8.2).

C10. Why is early minus late proportional to the error?

$R(\varepsilon - \frac{d}{2}) - R(\varepsilon + \frac{d}{2}) \approx -2R'(\frac{d}{2})\varepsilon$ (Section 8.1).

D. DLL vs MLE

D1. Why d = 0.5 chip?

Narrower spacing approximates the derivative better but shrinks the linear region; 0.5 is a standard compromise. Jitter ≈ 1.07 × CRLB.

D2. Where does K = 2/17 come from?

$N_{\text{eff}} = 16$ updates = 64 chips, the batch length: $K = 2/(N_{\text{eff}} + 1)$.

F: So "DLL ≈ CRLB" is by construction? — Yes, and we say so. A narrower loop goes below the 64-chip CRLB because it uses more data.

D3. Is the comparison fair?

In steady state yes; near threshold no — the DLL sees 640 chips. That is why we plot the ZZB for the DLL's own data, the only bound valid for it.

D4. Isn't the offset calibration b_j cheating?

It uses only the known pilot; with deliberately wrong timing the results were identical. Without it the DLL is biased (MSE ≈ 2 × CRLB at 30 dB).

D5. Which is better, DLL or MLE?

Above threshold equal (by design). Near threshold: same MSE threshold (−0.12 dB, CI [−0.87, 0.59]) but the DLL makes fewer gross errors (0.68 dB more robust) thanks to more data. Far below threshold the DLL is much worse (loss of lock).

D6. Why a first-order loop?

Constant delay, no Doppler: no steady-state error. With Doppler a second-order loop is needed.

E. Experiments

E1. How reliable are your numbers?

Up to 20,000 trials per point, bootstrap 95 % CIs, fixed seeds, tests against analytic facts, independent re-implementation, audit (Section 9).

E2. The 3 dB threshold definition looks arbitrary.

Fixed before running; a second, outlier-robust definition is also reported; we state conclusions that hold under both or say when they differ.

E3. The ZZB is 3.6 dB below your MLE threshold. Is the ZZB wrong?

Both limits match analytically and numerically, an independent implementation agrees to 0.12 %. The gap = 2.6 dB bound looseness + 0.85 dB MLE sub-optimality; it depends on the pilot (2.7–5.4 dB) and shrinks for a wider prior.

F: Mallat et al. found their bound much tighter. — Different signal (single pulse vs PN code with side-lobes) and definitions; no contradiction claimed.

E4. Why does the MLE's MSE exceed the prior variance at very low SNR?

The arg-max of a noise process piles up at the window edges (~20 % exactly on an edge): 3.09 vs 1.33 chips². The MMSE goes to 1.33.

E5. Why synthetic data only?

The bounds need the exact model and ground truth; real data add Doppler, multipath, bits and clock effects — future work.

F. Novelty (answer honestly)

F1. What is your own contribution?

The theory up to 70% is known. Ours: a carefully verified study for one model, plus observations not found stated for PN pilots — ZZB looseness against the MMSE (2.6 + 0.85 dB), its pilot dependence and prior-width trend, the DLL failure decomposition, the criterion-dependent DLL-vs-MLE result. The research contribution is the proposal.

F2. Hasn't the ZZB been applied to GNSS code tracking?

Yes — Emmanuele et al. (ICASSP 2012), Liu et al. (ION ITM 2016), for one integration period. We propose dynamics and loss of lock.

F3. Isn't loss of lock solved by MTLL theory?

MTLL gives exit times of a given loop. We found no work linking MTLL to an MSE bound or to the optimal tracker's threshold.

F4. How sure are you that it is novel?

Not sure — medium-low. Our first survey missed the Emmanuele papers, which is why we say "we found no prior work that…".

F5. Why not deep learning or RL?

Nothing requires it: the optimal tracker is computable exactly (1-D grid). Adaptive-bandwidth RL exists only as a proof of concept.

G. Future work

G1. What if H2 fails?

Then the DLL's near-threshold loss is partly jitter inflation inside lock — itself measurable with the Markov chain.

G2. Why a grid filter, not a particle filter?

One-dimensional state: a fine grid computes the Bayes recursion essentially exactly, no Monte Carlo error.

G3. Link to robotics / autonomous systems?

The DLL is a feedback estimator and the optimal tracker is the same Bayes filter used in robot localisation; predicting loss of lock is an integrity problem. Robot UWB data were rejected as forced: their outliers are NLOS, not thermal.

G4. How would this extend to a real receiver?

Add Doppler (2-D state, second-order loop), carrier phase, data bits, multipath; the grid filter stays feasible in 2-D.

A Notation and equation sheet

Symbol	Meaning
$\tau, \hat{\tau}, e = \hat{\tau} - \tau$	true delay, estimate, error (chips)
$N_c = 64$, $f_s = 4$	pilot length (chips), samples per chip
$\alpha = 0.5$	raised-cosine roll-off
β^2	mean-square (Gabor) bandwidth, 1/chip²
E/N_0, E_c/N_0	batch / per-chip energy-to-noise ratio
T_a	prior width (chips)
$\rho(h)$	normalised autocorrelation
d, K, B_L, N_{eff}	early–late spacing, loop gain, noise bandwidth, effective averaging length

Quantity	Formula
SNR	$E/N_0 = \sum s^2/(2\sigma^2)$
MLE	$\arg\max_\theta \sum r\, s(t-\theta) - \frac{1}{2}\sum s(t-\theta)^2$
CRLB	$1/(2(E/N_0)\beta^2)$, $\beta^2 = \sum s'^2/\sum s^2$
ZZB	$\int_0^{T_a} h\, \mathcal{V}\{(1-\frac{h}{T_a})Q(\sqrt{(E/N_0)(1-\rho(h))})\}dh$
ZZB limits	$T_a^2/12$ (no signal); CRLB (strong signal)
DLL	$\theta_{j+1} = \theta_j - K(E-L-b_j)/\bar{k}$; $N_{\text{eff}} = (2-K)/K$; $B_L T = K/(2(2-K))$
PCRB (proposal)	$J_k = J_D + 1/(q + 1/J_{k-1})$; $P_\infty = \frac{1}{2}(-q + \sqrt{q^2+4qR})$

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